AS#2 Process Capability and SPC Case Study

Craft Beer brewery in Indiranagar, India

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Introduction

- ## Background

Factory-produced beer

vs

Craft beer

  - Unlike factory-produced beer, craft beer is produced in small breweries.
    - There is also a craft beer restaurant within PNU's commercial district! ☺
  - However, due to capital limitations, small breweries cannot strictly control their quality compared to factory-produced beer.

- ## Case study topic
  - Process data of Indian craft beer brewery "TOIT" could be collected from Kaggle.
  - **Our goal** : Performing the process capability analysis of TOIT Brewery and implement the SPC mechanism.

kaggle

  - We would like to collect data from the process of craft beer, construct a control chart using Minitab software, and analyze the process capability of craft beer.

Data gathering

▪ Data gathering process

Raw data from kaggle

Beer_Style	SKU	Location	Fermentation_Time	Temperature	pH_Level	Gravity	Alcohol_Content	Bitterness	Color	Ingredient_Ratio
Wheat Beer	Kegs	Whitefield	16	24.204251	5.289845	1.039504	5.370842	20	5	1:0.32:0.16
Sour	Kegs	Whitefield	13	18.086763	5.275643	1.059819	5.096053	36	14	1:0.39:0.24
Wheat Beer	Kegs	Malleswaram	12	15.539333	4.778016	1.037476	4.824737	30	10	1:0.35:0.16
Ale	Kegs	Rajajinagar	17	16.418489	5.345261	1.052431	5.509243	48	18	1:0.35:0.15

Main column

Temparture : average temperature during fermentation

Fermantation_time : total duration of fermentation

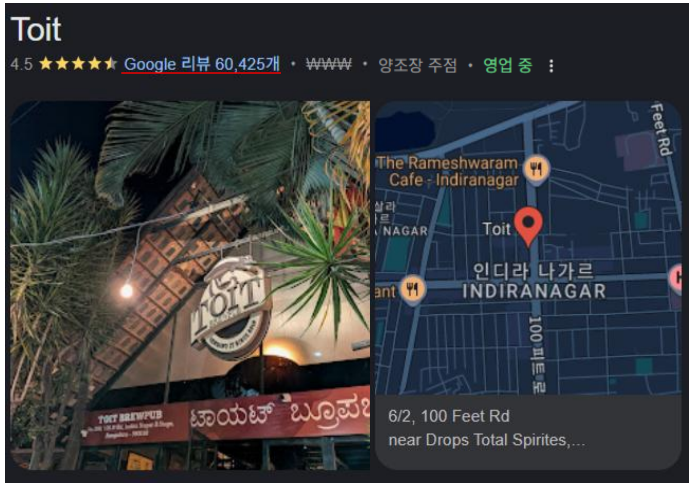
Ingredient_Ratio : ratio of key ingrediants(malt:hop:..)

Alcohol_content : final alcohol percentage

Bitterness : International Bitterness Units

Gravity : Amount of wort (malt) before fermentation

▪ Data Processing procedure



Toit Bangalore - Indiranagar

① Select the most famous breweries based on the combination of beer style and location

② Data extraction for same beers using Color and Ingredient Ratio

Batch	Beer_Style	SKU	Location	Fermentation_Time	Temperature	pH_Level	Gravity	Alcohol_Content	Bitterness	Color	Ingredient_Ratio
1	Ale	Pints	Indiranagar	14	23.90279282	4.527526916	1.049763139	5.851573944	33	17	1:0.44:0.16
1	Ale	Bottles	Indiranagar	17	19.89149729	5.499371314	1.067128668	5.655957438	50	17	1:0.43:0.15
1	Ale	Bottles	Indiranagar	12	16.67131879	4.775565537	1.032901397	5.647359557	36	17	1:0.42:0.20
2	Ale	Pints	Indiranagar	15	18.2157106	5.005422665	1.079359471	4.635655935	56	17	1:0.45:0.23
2	Ale	Pints	Indiranagar	16	21.22537895	5.140458413	1.032039528	4.880870351	42	17	1:0.42:0.24
2	Ale	Kegs	Indiranagar	17	20.39125939	4.868929945	1.069377575	5.529248261	25	17	1:0.35:0.15

③ Samples with three data point produced on the same day are extracted (1~2 week)
→ # of sample : 20 / sample size : 3

▪ Additional information

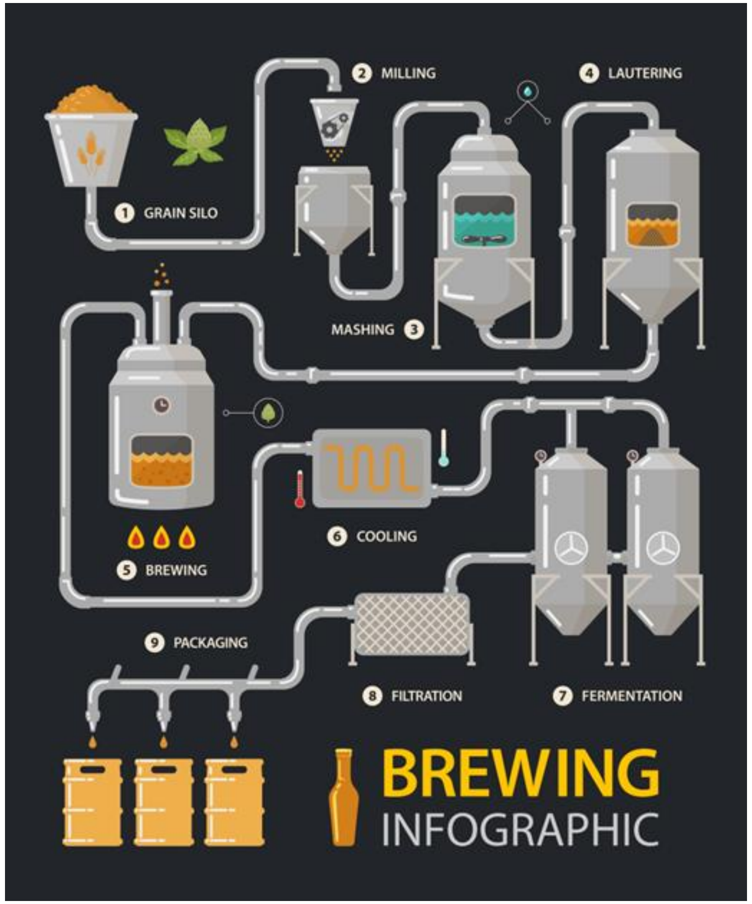
Beer Style Guidelines

AND

Beer-discovery application

Production Process

- **Beer production process**



[1]

Grain silo & Milling(원재료 가공) : **Ingredient Ratio** are determined

Mashing(담금) : **pH level** and **Gravity**(초기 농도) are determined, and saccharification is performed

Lautering(여과) : Securing clear **wort**(맥즙) and **volume**

Brewing(끓임) : **Color** and **Bitterness** determined

Cooling(냉각) – **Fermentation** conditions **temperature** setting, fermentation start

Fermentation(발효) : **Gravity** reduction, **pH level** change, **Alcohol content** creation

Filteration(여과) – **Color** tidying, some **loss** of amount

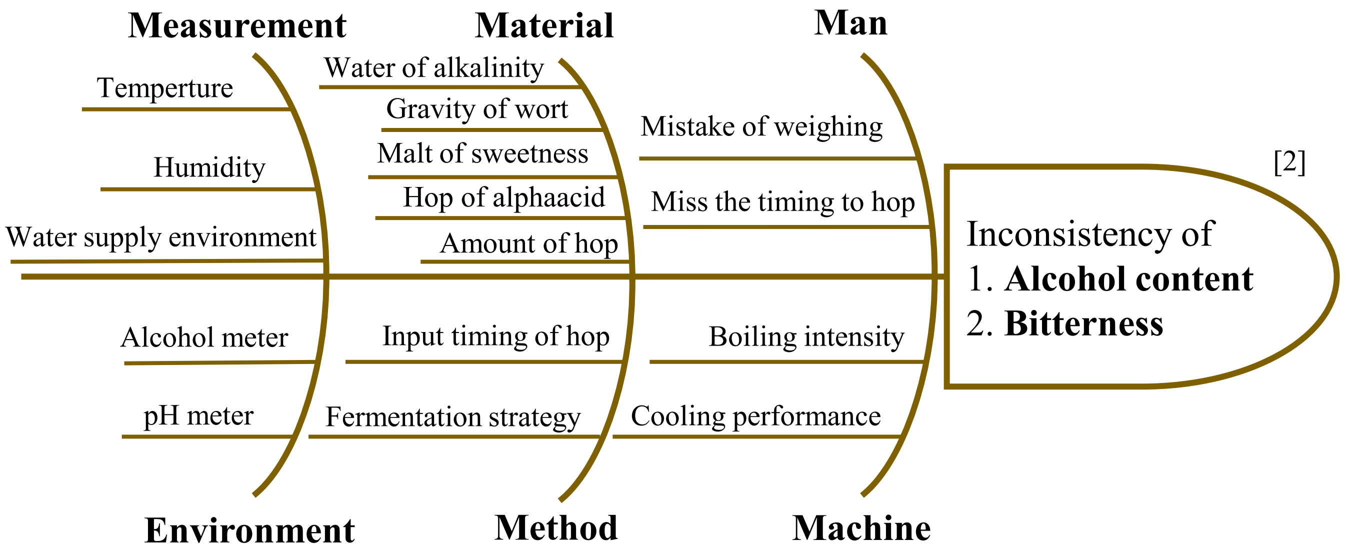
Packaging(포장) – **SKU** to complete the final product

Our Focusing Quality Characteristics - direct effect on **taste**

[1] Archana, G., Gaurav, S., Shruti, M., Sarita, M., Anuradha, S., & Kumar, D. M. (2024). BEER PRODUCTION BY FERMENTATION PROCESS: A REVIEW. *Journal of Microbiology, Biotechnology & Food Sciences*, *13*(4)

Cause and Effect

- **Cause and Effect Diagram**



[2]

1. Alcohol content
   - N-type
   - The alcohol content of a typical beer is 4.0 to 6.0 degrees [3]
2. Bitterness
   - N-type
   - The bitterness of a typical beer is 20 to 40 [4]

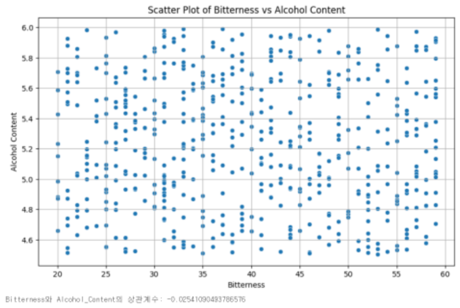
– It is important to balance alcohol content and bitterness

- **Why important Alcohol content & Bitterness?**
  - A direct impact on the taste and style of beer

  - Alcohol content ↔ ABV
  - Bitterness ↔ IBU
  - Determined at different process stages
    - ABV: Determined by grain/sugarization/fermentation
    - IBU : Determined by inputting and boiling hops

  - Correlation = **-0.02**

→ **There is no correlation**



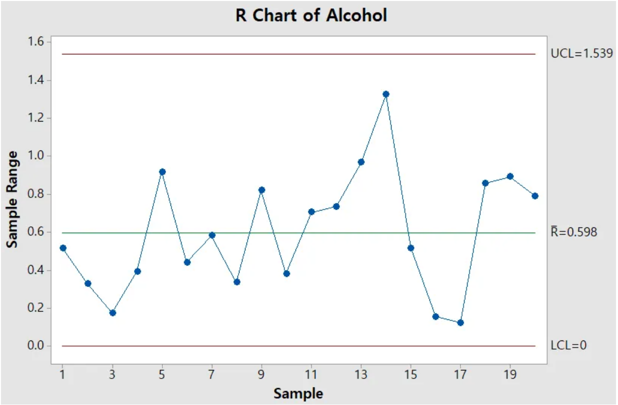
[2] Bamforth, C. W. (2011). 125th Anniversary Review: The Science and Understanding of the Flavor of Beer. Journal of the Institute of Brewing, 117(4), 488-497.

[3] Godínez-Hernández, C. I., Rodríguez-Cardona, T. D. J., Rendón-Huerta, J. A., Cervantes-Paz, B., & Michel-Cuello, C. (2025). Physicochemical Kinetics and Determination of Methyl and Ethyl Alcohols in Own-Manufactured Craft Beer and Comparison with Commercial Mexican CraftBeers. *Beverages*, *11*(1), 28.

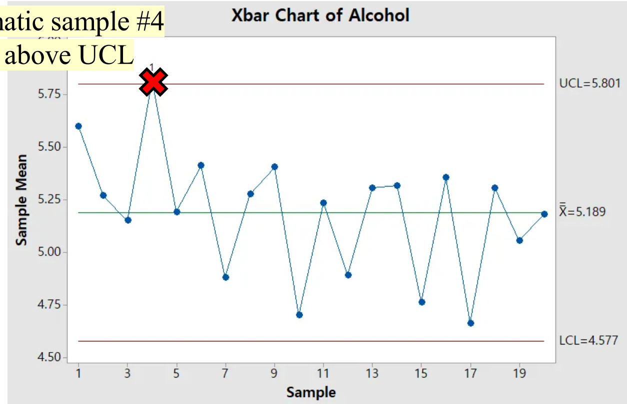
[4] https://www.brewersfriend.com/2009/01/24/beer-styles-ibu-chart-graph-bitterness-range/

Construction R & $\bar{X}$ chart

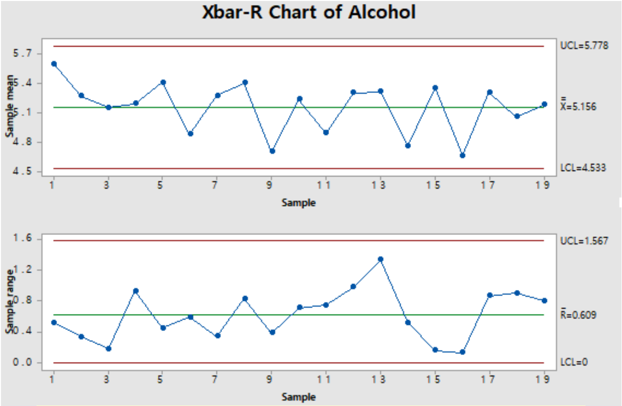
▪ Alcohol Content



Problematic sample #4
Rule 1 : above UCL



Repeat the same process, confirming that there is no out of control



All cleared X bar, R chart

▪ How to find special cause

Data table of sample #4

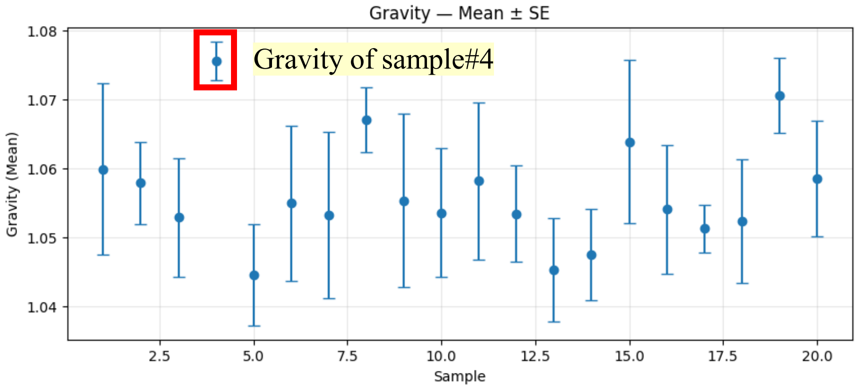
Beer_Style	Fermentation_Time	Temperature	pH_Level	Gravity	Alcohol_Content	Bitterness	Color	Ingredient_Ratio
Ale	14	15.18050297	5.013351502	1.079597292	5.59668163	40	17	1:0.36:0.24
Ale	11	20.07400084	5.395488652	1.076980296	5.871778751	36	17	1:0.43:0.17
Ale	10	16.7184682	5.469574493	1.070086692	5.990814547	33	17	1:0.36:0.22

※ Statistical analysis such as ANOVA was not performed because the number of observations per sample was too small as n=3, so the power was low

Gravity(맥즙의 양) : Specific gravity of the molt before fermutation

By domain knowledge, [5]

Cause Gravity	→	**Effect** Alcohol content



● : **Average** of fermentation time

: ±***SE*** of fermentation time

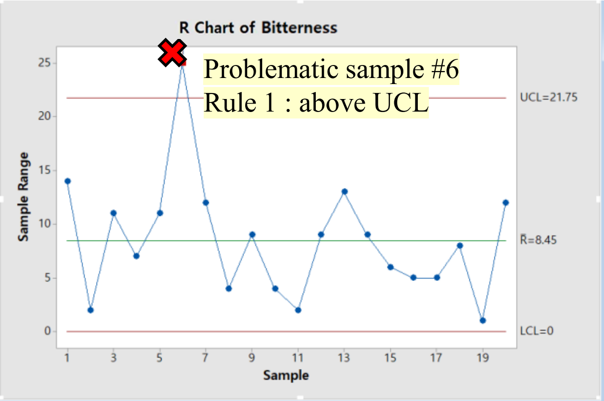
Special cause : Material !

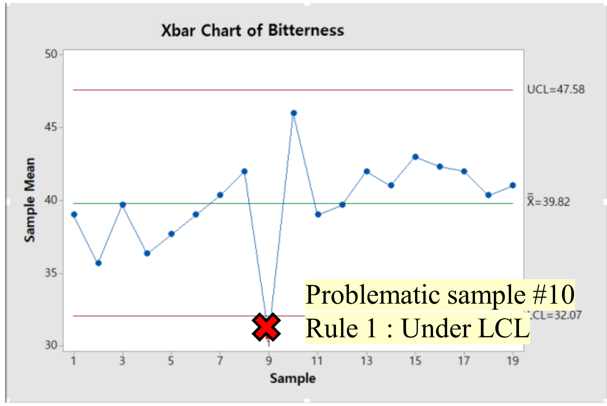
Sample #4 is **highest average Gravity**.
The sugar content in the wort increases the alcohol content

[5] Bamforth, C. W. (2000). Perceptions of beer foam. *Journal of the Institute of Brewing*, *106*(4), 229-238.

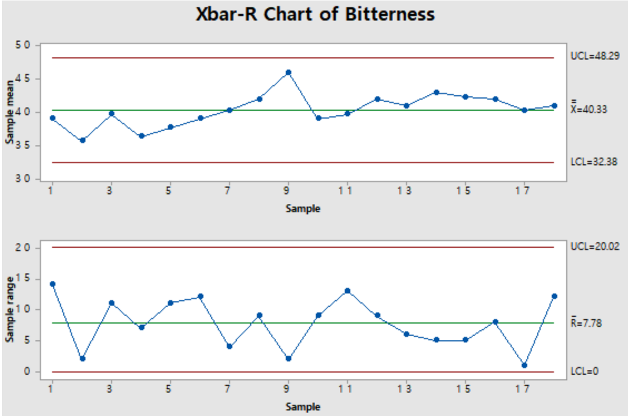
Construction R & $\bar{X}$ chart

▪ Bitterness of beer





Repeat the same process, confirming that there is no out of control



All cleared X bar, R chart

▪ How to find special cause

In page 2, beer production process… [6]

- ✓ **Brewing(끓임)** : Color and **Bitterness** determined [7]
- ✓ It is known that the amount of **malt&hops** directly affects **bitterness**.

● : **Average** of fermentation time

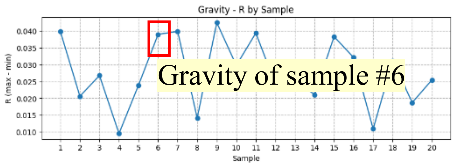
: ±***SE*** of fermentation time

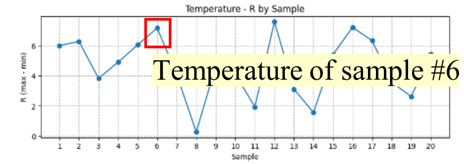
Data table of sample #6

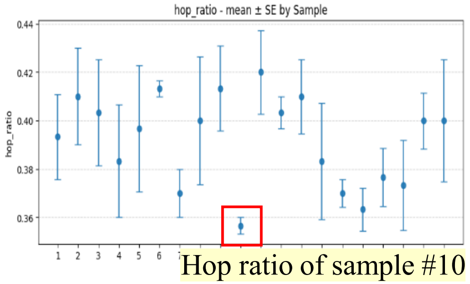
Beer_Style	Fermentation_Time	**Temperature**	pH_Level	**Gravity**	Alcohol_Content	**Bitterness**	Color	**Ingredient_Ratio**
Ale	12	22.03593	4.914212	1.0743162	5.295355	30	17	1:0.42:0.25
Ale	11	23.31229	5.043512	1.055225255	5.690288	49	17	1:0.41:0.23
Ale	19	16.15107	5.22493	1.035315115	5.249454	55	17	1:0.41:0.19

Data table of sample #10

Beer_Style	Fermentation_Time	**Temperature**	pH_Level	**Gravity**	Alcohol_Content	**Bitterness**	Color	**Ingredient_Ratio**
Ale	18	19.74094534	4.909863223	1.072164287	4.549884952	32	17	1 : 0.35 : 0.15
Ale	10	19.38270188	5.203749796	1.042387411	4.617606584	32	17	1 : 0.36 : 0.22
Ale	18	23.48962445	4.944733335	1.046120381	4.931285768	28	17	1 : 0.36 : 0.20







Sample #6 has a **higher R** than other samples in **Gravity and Temperature** at the same time.

Special cause : Environment&Material !

Sample #10 has the **lowest average of hop ratio to malt.**

Special cause : Material !

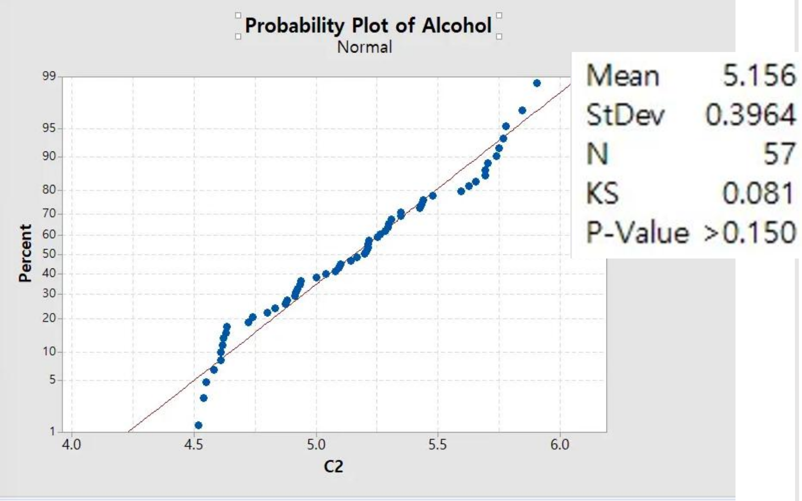
[6] Kappler, S., Krahl, M., Geissinger, C., Becker, T., & Krottenthaler, M. (2010). Degradation of iso-α-acids during wort boiling. Journal of the Institute of Brewing, 116(4), 332-338.
[7] Jaskula, B., Kafarski, P., Aerts, G., & De Cooman, L. (2008). A kinetic study on the isomerization of hop α-acids. Journal of agricultural and food chemistry, 56(15), 6408-6415

Normality Test : Validation for Process Capability Analysis

- **Alcohol content**

↓	C1-T	C2
	subscript	
1	C4	5.77720
2	C4	5.29200
3	C4	5.03777
4	C4	4.99845
5	C4	5.29536
6	C4	4.62940
7	C4	5.13945
8	C4	5.65164
9	C4	4.54988
10	C4	4.91664
11	C4	4.60900
12	C4	5.07847
13	C4	5.59135
14	C4	5.09752
15	C4	5.43649
16	C4	4.63363

Merged data of Alcohol



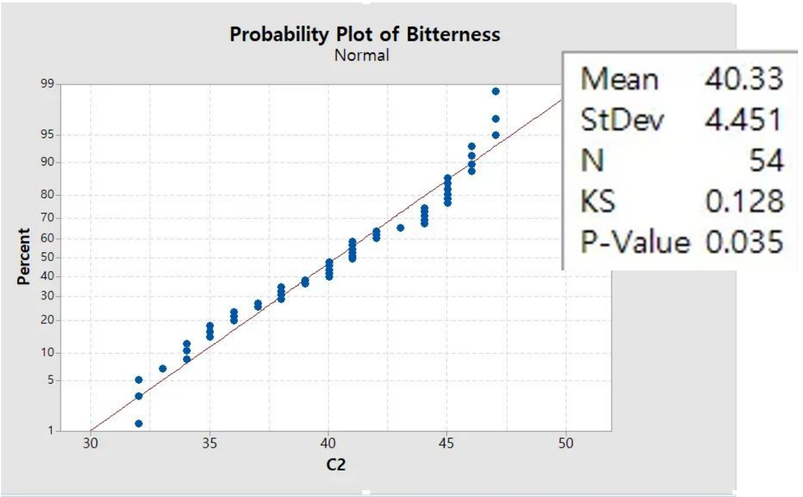
Accept H_0 because p-value > 0.01

H_0: Data follow normal distribution
H_1: Data doesn't follow normal distribution

- **Bitterness**

↓	C1-T	C2
	subscript	
1	C1	39
2	C1	35
3	C1	40
4	C1	36
5	C1	34
6	C1	41
7	C1	41
8	C1	43
9	C1	45
10	C1	36
11	C1	42
12	C1	41
13	C1	41
14	C1	44
15	C1	44
16	C1	42

Merged data of Bitterness



Accept H_0 because p-value > 0.01

Result : The assumption of normality is satisfied.

Why level of significance $\alpha = 0.01$, not 0.05? [8]
- In biological processes like fermentation, natural variations exist.
- Using α = 0.01 prevents excessive rejection of normality, allowing us to proceed with capability analysis.

[8] Mudge, J. F., Baker, L. F., Edge, C. B., & Houlahan, J. E. (2012). Setting an optimal α that minimizes errors in null hypothesis significance tests. *PloS one*, *7*(2), e32734.

Process Capability Analysis

- **Process capability**

USL and LSL for Alcohol and Bitterness were referenced from the Beer Style Guidelines
Alcohol content : LSL : 4.50% / USL : 7.10%
Bitterness : LSL : 35 / USL : 63

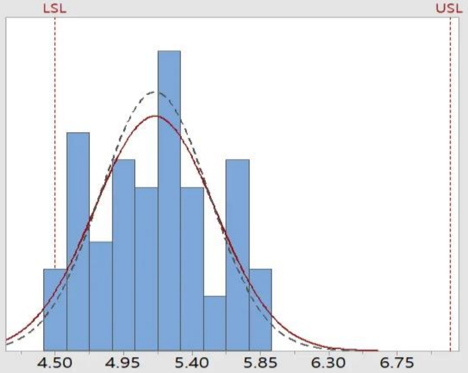
Brewers Association
2025 Beer Style Guidelines

Alcohol

LSL 4.50 % — USL 7.10 %

Process Capability Report for Alcohol

Process Data	
LSL	4.5
Target	*
USL	7.1
Sample Mean	5.15556
Sample N	57
StDev(Overall)	0.396371
StDev(Within)	0.359485



Overall Capability	
Pp	1.09
PPL	0.55
PPU	1.64
Ppk	0.55
Cpm	*

Potential (Within) Capability	
Cp	1.21
CPL	0.61
CPU	1.80
Cpk	0.61

Performance	Observed	Expected Overall	Expected Within
PPM < LSL	0.00	49072.85	34105.24
PPM > USL	0.00	0.47	0.03
PPM Total	0.00	49073.31	34105.27

[9]

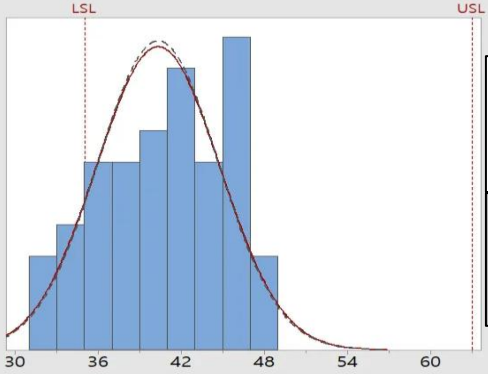
For **Alcohol**, C_p is greater than P_p, indicating that the σ_{within} is smaller than the $\sigma_{overall}$. This reflects substantial batch-to-batch variation, meaning that long-term process performance is less stable than short-term behavior.

Bitterness

LSL 35 (IBU) — USL 63 (IBU)

Process Capability Report for Bitterness

Process Data	
LSL	35
Target	*
USL	63
Sample Mean	40.3333
Sample N	54
StDev(Overall)	4.45099
StDev(Within)	4.36783



Overall Capability	
Pp	1.05
PPL	0.40
PPU	1.70
Ppk	0.40
Cpm	*

Potential (Within) Capability	
Cp	1.07
CPL	0.41
CPU	1.73
Cpk	0.41

Performance	Observed	Expected Overall	Expected Within
PPM < LSL	129629.63	115412.77	111033.67
PPM > USL	0.00	0.18	0.11
PPM Total	129629.63	115412.94	111033.78

For **Bitterness**, C_p and P_p are nearly identical, meaning that σ_{within} and $\sigma_{overall}$ are almost the same. This suggests minimal batch-to-batch variation, and the long-term quality level is similar to the short-term level.

[9] Steiner, S., Abraham, B., & MacKay, J. (1997). Understanding process capability indices. *Waterloo, Ontario*.

Conclusions and Recommendations

Conclusion

– Control chart

1. Through Experience based problem define, we want to manage the quality of Beer Taste
   - ✓ Conducted root cause analysis by categorizing data columns based on manufacturing process stages
   - ✓ From the domain knowledge,we can get ABV(Alcohol contern)and IBU(Bitterness) is the key factors to Taste of beer
2. From the data , We can found <u>assingnable problem</u> from Alcohol content and Bitterness
3. By removing this data, it was shown that all processes were <u>in control</u>.

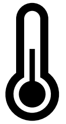
– Process capability analysis

1. There is a <u>mean shift</u> in both alcohol content and bitterness.
2. In bitterness, there is little difference between $\boldsymbol{C_p}$ and $\boldsymbol{P_p}$, and between $\boldsymbol{C_{pk}}$ and $\boldsymbol{P_{pk}}$, so it seems to be consistent with the process.
3. In alcohol content, there is a difference between the two indicators. This suggests that there is additional variability, not only in the process itself.

Recommendation


홉(hop)


맥아(malt)


…

– Control chart

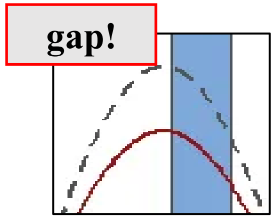
1. Maintain optimal gravity(맥즙 초기 농도) :
   The sugar content in the wort(맥즙) increases the alcohol content
   - ✓ Suggest improvement
     - → Introducing a device that will keep the amount of wort(맥즙) constant
2. Maintain an adequate hop-to-malt ratio :
   A lower hop-to-malt ratio leads to increased bitterness
   - ✓ Suggest improvement
     - → Implement an Automated Dosing System to optimize the hop-to-malt ratio
3. Minimize process variability :
   maintaining consistent wort input and temperature
   - ✓ Suggest improvement
     - → Improve filter to maintain the initial concentration determined in the mashing step is not diluted or changed
     - → Add Automatic Temperature Control to strictly maintain the process temperature within 20 °C- 22 °C

– Process capability analysis

1. We confirmed a mean shift. Therefore, we need to apply DMAIC for systematic improvement, specifically by revisiting the DoE stage to fix the shift.
2. Alcohol content involves biological processes. We suspect the variation is caused by this, but we can't be sure yet. So, we need to collect and analyze more data.

Lessons Learned

- **Get a deeper understanding of process capability analysis and SPC mechanism**
  - We were able to have a deep understanding of the reasons for using the control chart and process capability.
  - By looking at the results of the process capability analysis using Minitab, it was possible to know the meaning and analysis method of the indicators(P_p, P_{pk}) that were not learned in class.



- **The Importance of Data Collecting**
  - Due to the characteristic of craft beer, mass production was not performed, and the size of the data compared to the large-scale process was small due to capital limitations.
  - Due to the characteristic of the craft beer process, the sampling cycle is long, from 1 to 2 weeks, and due to this, the change in the control chart could not be observed in detail.
  - Statistical and additional analysis could not be performed due to the limitations of data size.

- **Limitations**
  - The control chart showed the Special Cause, but the data didn't clearly show the underlying factor. This taught us that improvement requires conducting experiments and analysis several times.
  - When looking for special reasons in a situation where data is scarce, we learned to apply Rule of Eyes.[10]
    (But it was not fitted for our case. ☹)
  - We learned it difficult to guess the reason only by looking at the data without experiencing the actual beer manufacturing process.

[10] Cumming, G., & Finch, S. (2005). Inference by eye: confidence intervals and how to read pictures of data. American psychologist, 60(2), 170.